Acute Kidney Injury (AKI)

All key points for the exam

1. Definition (KDIGO)

AKI is diagnosed if **any one** of the following is present:

Criterion	Value
Rise in serum creatinine (SCr)	**≥ 0.3 mg/dL** within **48 hours**
Or rise in SCr	**≥ 1.5 × baseline** within **7 days**
Or low urine output	**< 0.5 mL/kg/h** for **6 hours**

KDIGO Staging

Stage	Serum creatinine	Urine output
1	**1.5–1.9 × baseline**, or rise **≥ 0.3 mg/dL**	< 0.5 mL/kg/h for **6–12 h**
2	**2.0–2.9 × baseline**	< 0.5 mL/kg/h for **≥ 12 h**
3	**≥ 3 × baseline**, **or** SCr **≥ 4 mg/dL**, **or** start of **renal replacement therapy (dialysis)**	< **0.3** mL/kg/h for **≥ 24 h**, **or anuria** for **≥ 12 h**

Note: Older classification systems that may appear in questions: **RIFLE** (Risk, Injury, Failure, Loss, End-stage) and **AKIN**.

Urine output terms

- **Oliguria:** < **400 mL/day**.
- **Anuria:** < **50–100 mL/day**.
- **Creatinine rises late**, so early on it **does not reflect the true kidney function**.

2. Types of AKI by Cause

Type	Mechanism	Examples
Prerenal (most common)	**Reduced blood flow** to the kidney; the kidney itself is intact	Dehydration, bleeding, vomiting and diarrhea, **heart failure**, **cirrhosis**, **drugs** (NSAIDs, ACEi/ARB)
Intrinsic	**Damage to the kidney itself**	**ATN** (most common), **AIN**, glomerulonephritis (GN), vascular causes
Postrenal	**Obstruction** of the urinary tract	**BPH**, stones, tumors, **drug crystals**, **anticholinergic** drugs

ATN = Acute Tubular Necrosis • **AIN** = Acute Interstitial Nephritis

3. Differentiating by Lab Tests (very important for the exam)

Test	Prerenal	ATN (intrinsic)
BUN:SCr ratio	**> 20**	**< 10–15**
FENa	**< 1%**	**> 2%**
FEUrea (if on diuretics)	**< 35%**	**> 50%**
Urine sodium (UNa)	**< 20** mEq/L	**> 40** mEq/L
Urine osmolality	**> 500** mOsm/kg (concentrated urine)	**< 350** (close to plasma)
Specific gravity	**> 1.020**	About **1.010**

Test	Prerenal	ATN (intrinsic)
Urine sediment	**Hyaline casts** or normal	**Muddy brown granular casts**

Formulas

FENa (%) = (Urine Na × Serum Cr) ÷ (Serum Na × Urine Cr) × 100

FEUrea (%) = (Urine urea × Serum Cr) ÷ (Serum urea × Urine Cr) × 100

Why is FENa low in prerenal AKI? Because the kidney is **intact** and is trying to **retain sodium and water**. In **ATN**, the tubules are **damaged** and cannot retain sodium.
Diuretics falsely raise FENa, so **use FEUrea** instead.

Urine sediment by cause

Sediment	Diagnosis
Hyaline casts	Prerenal
Muddy brown granular casts	**ATN**
WBC casts + **eosinophiluria**	**AIN**
RBC casts	**Glomerulonephritis (GN)**
Crystals	Crystal obstruction (e.g., acyclovir)
Dipstick positive for blood, but no RBCs on microscopy	**Rhabdomyolysis** (due to myoglobin)

4. Drugs That Cause AKI (most important section for pharmacists)

A) Drugs causing prerenal (hemodynamic) AKI

Drug	Mechanism
NSAIDs (including COX-2 inhibitors)	**Afferent arteriole vasoconstriction**, because they block vasodilating prostaglandins
ACEi / ARB (and aliskiren)	**Efferent arteriole vasodilation**, because they block angiotensin II
Cyclosporine / Tacrolimus	**Afferent arteriole vasoconstriction**
Diuretics	**Volume depletion**
Contrast media	**Vasoconstriction**, plus **direct toxicity (ATN)**

Triple whammy: NSAID + ACEi or ARB + diuretic → very high AKI risk, especially in the elderly.

ACEi and ARB

- A **creatinine rise of up to 30%** after starting the drug is **acceptable**; do not stop it.
- **More than 30%** → stop the drug and look for **renal artery stenosis**.
- **Contraindicated** in **bilateral renal artery stenosis**.

B) Drugs causing ATN (direct tubular toxicity)

Drug	Key points
Aminoglycosides (gentamicin, tobramycin, amikacin)	Appears **after 5–7 days** • Usually **non-oliguric** • **Hypomagnesemia and hypokalemia** • **Extended-interval (once-daily) dosing** is less toxic • **Monitor troughs** • Also cause **ototoxicity**
Vancomycin	Higher risk with **trough > 15–20** • **AUC-guided monitoring (400–600)** is less nephrotoxic than trough-guided monitoring • **With piperacillin-tazobactam → higher AKI risk**
Amphotericin B	**Hypokalemia and hypomagnesemia** • **Renal tubular acidosis (RTA type 1)** • **Prevention: 500 mL normal saline** before and after the dose (**saline loading**) • **Liposomal formulation** is less toxic
Cisplatin	**Hypomagnesemia** (very characteristic) • **Prevention:** aggressive saline hydration + magnesium • **Amifostine** is protective
Contrast media	Details in section 6
Foscarnet, Cidofovir	**Cidofovir:** given with **probenecid** + hydration • **Foscarnet:** calcium and magnesium disturbances
Tenofovir (TDF)	**Fanconi syndrome** (glucose, phosphate and bicarbonate wasting in urine) • **TAF** is less nephrotoxic
IVIG (sucrose-containing)	**Osmotic nephrosis**
Mannitol (high doses)	Osmotic nephrosis
Zoledronic acid	ATN; adjust dose by CrCl
Polymyxin B / Colistin	**High** nephrotoxicity

C) Drugs causing AIN (allergic interstitial nephritis)

Classic triad: Fever + rash + eosinophilia. All three appear together in only **about 10%** of cases.

Class	Examples
Beta-lactams (most common)	**Methicillin** (classic), **penicillins**, **cephalosporins**
Sulfonamides	**TMP-SMX**, **furosemide**, thiazides
Rifampin	
Fluoroquinolones	Ciprofloxacin
Proton pump inhibitors (PPIs)	Omeprazole and others (very important because they are so common)
NSAIDs	May occur **without** fever and rash, and with **proteinuria (nephrotic)**
Others	**Allopurinol**, phenytoin

Treatment: Stop the offending drug → consider **corticosteroids** if no improvement.

D) Drugs causing postrenal (obstructive) AKI

Drug	Mechanism	Prevention
IV acyclovir	**Crystals**	**Adequate hydration**, **slow infusion** (over ≥ 1 hour)
High-dose methotrexate	Crystals	Hydration + **urine alkalinization** with bicarbonate + **leucovorin** + **glucarpidase** for toxicity
Sulfonamides (sulfadiazine)	Crystals	Hydration
Indinavir	Crystals and stones	Drink at least **1.5 L of water** daily
Triamterene	Stones	
Anticholinergics, **opioids**	**Urinary retention**	Especially with **BPH**
Tumor lysis syndrome	**Uric acid crystals**	Hydration + **allopurinol** or **rasburicase** (**contraindicated in G6PD deficiency**)

E) Drugs causing AKI by other mechanisms

Mechanism	Drugs
Rhabdomyolysis	**Statins** (especially with **gemfibrozil** or **CYP3A4 inhibitors**), daptomycin, **cocaine**, alcohol
Thrombotic microangiopathy (TMA / HUS / TTP)	**Cyclosporine**, **tacrolimus**, **gemcitabine**, **clopidogrel** and **ticlopidine**, **quinine**, **VEGF inhibitors** (bevacizumab)
Glomerulonephritis	**Hydralazine** (ANCA vasculitis), **penicillamine**, **gold**, **NSAIDs** (minimal change disease), **lithium**
Lithium	**Nephrogenic diabetes insipidus** + **chronic kidney disease**

F) Pseudo-AKI (false rise in creatinine) — classic trap

Creatinine **rises** but **GFR is unchanged**, because these drugs **block tubular secretion of creatinine**:

- **Trimethoprim** (in TMP-SMX; also causes **hyperkalemia**)
- **Cimetidine**
- **Dolutegravir**, **bictegravir**
- **Cobicistat**, **ritonavir**
- **Fenofibrate**
- **Cefoxitin** and **flucytosine** interfere with the **creatinine assay**

Rise in BUN without kidney injury

Corticosteroids, **tetracyclines** (protein catabolism), **GI bleeding**, high-protein **TPN**.

5. Risk Factors for Drug-Induced AKI

- **Elderly**.
- Pre-existing **chronic kidney disease (CKD)**.
- **Dehydration** or **volume depletion**.
- **Diabetes**.
- **Heart failure**, **cirrhosis**.
- **Sepsis**.
- **Combining several nephrotoxic drugs**.
- **High doses** or **long duration**.

6. Contrast-Associated AKI

Point	Details
Onset	**Within 24–72 hours**, peaks at **3–5 days**
Risk factors	**eGFR < 30** (most important), diabetes, heart failure, dehydration, elderly, **large contrast volume**
Prevention	**IV isotonic saline** (e.g., **1 mL/kg/h** for **6–12 h** before and after the procedure) • Use the **lowest contrast volume** • **Low- or iso-osmolar** contrast • **Hold** NSAIDs and diuretics temporarily
Not beneficial	**N-acetylcysteine** (no proven benefit); **bicarbonate** is not better than saline
Metformin	**Hold** on the day of the procedure if **eGFR < 30**, **AKI**, or **intra-arterial contrast**; then **recheck renal function after 48 hours** before restarting (risk of **lactic acidosis**)

7. Management

General principles

> **No drug treats AKI or speeds recovery.** Management is **supportive**.

1. **Treat the cause**.
2. **Stop nephrotoxic drugs**.
3. **Manage fluids**.
4. **Treat complications**.
5. **Adjust drug doses**.

Fluids

- **Prerenal AKI with dehydration: isotonic crystalloids**.
- **Balanced crystalloids** (**Lactated Ringer's** or **Plasma-Lyte**) are **often preferred** over large volumes of **normal saline (NS)**, because NS causes **hyperchloremic metabolic acidosis**.
- **Avoid hydroxyethyl starch (HES)**: it **increases AKI and mortality**.
- **Albumin:** in **cirrhosis** and **hepatorenal syndrome**.

Loop diuretics

- Used **only to treat volume overload**.
- They **do not prevent AKI**, **do not treat it**, and **do not improve outcomes**. They only convert oliguria to higher urine output without improving kidney function.
- **High doses** may be needed in AKI. **Very high furosemide doses** cause **ototoxicity**.

Not recommended

- **Low-dose ("renal-dose") dopamine**: no benefit.
- **Fenoldopam**: no clear benefit.
- Routine **mannitol**.
- **N-acetylcysteine**.

Hepatorenal syndrome

- **Terlipressin + albumin** (preferred).
- **Or norepinephrine + albumin** in the ICU.
- **Or midodrine + octreotide + albumin**.

8. Managing Complications

A) Hyperkalemia (most dangerous)

Goal	Drug	Onset	Notes
1. Protect the heart	**IV calcium gluconate**	**Minutes**	**Does not lower potassium**; only stabilizes the cardiac cell membrane • Give **first** if there are **ECG changes**
2. Shift potassium into cells	**Regular insulin (10 units) + dextrose**	15–30 min	Monitor for **hypoglycemia**
	Nebulized albuterol (high dose)	15–30 min	
	Sodium bicarbonate		Only **with acidosis**
3. Remove potassium from the body	**Furosemide**		If the patient is still producing urine
	Sodium zirconium cyclosilicate (Lokelma)	About 1 hour	
	Patiromer (Veltassa)	Hours	Binds other drugs; **separate by 3 hours**
	Sodium polystyrene sulfonate (Kayexalate)	Hours	Risk of **intestinal necrosis**
	Dialysis		Most effective

Drugs that raise potassium (stop them): ACEi and ARB, **spironolactone**, **trimethoprim**, NSAIDs, heparin, tacrolimus, potassium supplements.

B) Metabolic acidosis

Sodium bicarbonate if **pH < 7.2** or **bicarbonate < 15**.

C) Hyperphosphatemia

Phosphate binders with meals: **calcium acetate**, **sevelamer**, **lanthanum**.

D) Volume overload

Fluid restriction + **loop diuretics** + **dialysis** if not responding.

9. Indications for Dialysis: AEIOU

Letter	Indication
A	**Acidosis**: severe and refractory to treatment
E	**Electrolytes**: severe, refractory **hyperkalemia**
I	**Intoxication**: dialyzable toxins (**lithium**, **methanol**, **ethylene glycol**, **salicylates**, theophylline, **metformin**)
O	**Overload**: volume overload refractory to diuretics (pulmonary edema)
U	**Uremia**: uremic complications (**pericarditis**, **encephalopathy**, **bleeding**)

Types of dialysis in AKI

Type	Use
Intermittent hemodialysis (IHD)	**Hemodynamically stable** patient
Continuous renal replacement therapy (CRRT), e.g., **CVVH**, **CVVHD**, **CVVHDF**	**Hemodynamically unstable**, hypotensive patient (ICU)
SLED (sustained low-efficiency dialysis)	In between

10. Drug Dosing in AKI (very important for pharmacists)

The problem with estimating kidney function

- **Cockcroft-Gault and eGFR equations are inaccurate in AKI**, because **creatinine is not at steady state**.
- If **creatinine is rising rapidly** or the patient is **oliguric** → **assume CrCl < 10 mL/min**.

Cockcroft-Gault equation

CrCl = [(140 − age) × weight (kg)] ÷ [72 × SCr] → × 0.85 for females

Weight used	When
Total body weight (TBW)	If **less than IBW**
Ideal body weight (IBW)	Normal weight
Adjusted body weight (AdjBW)	If **TBW > 120% of IBW** (obesity)

- **IBW (male)** = 50 + (2.3 × each inch over 5 feet)
- **IBW (female)** = 45.5 + (2.3 × each inch over 5 feet)
- **AdjBW** = IBW + 0.4 × (TBW − IBW)

Dose adjustment rules

Rule	Details
Loading dose	**Unchanged**, because it depends on **volume of distribution (Vd)**, not on the kidney
Maintenance dose	**Reduce the dose** or **extend the dosing interval**
Antibiotics in the first days	**Do not reduce doses too quickly**: underdosing may cause treatment failure, and many AKI cases improve within 48 hours
Therapeutic drug monitoring (TDM)	Vancomycin, aminoglycosides

Drugs removed by dialysis

A drug is **removed by dialysis** if it has:

- **Low molecular weight**.
- **Low protein binding**.
- **Small volume of distribution (Vd)**.
- **Water solubility**.

Dialyzable drugs are given after the dialysis session.
In CRRT, removal is **continuous**, so doses are **higher** than in intermittent HD and depend on the **effluent rate**.

Drugs to avoid or adjust in renal failure

Drug	Reason
Metformin	**Lactic acidosis**: contraindicated if **eGFR < 30**
Meperidine, **morphine**, **codeine**	**Metabolite accumulation** (seizures, excessive sedation). Alternatives: **hydromorphone** or **fentanyl**
Enoxaparin	**CrCl < 30 → 1 mg/kg once daily** instead of twice daily
Nitrofurantoin	Ineffective and toxic if **CrCl < 30**
Gabapentin, **pregabalin**	**Accumulation** → excessive sedation
Glyburide	Prolonged **hypoglycemia**. Alternative: **glipizide**
Direct oral anticoagulants (DOACs)	Adjust by **CrCl**
Magnesium- and phosphate-containing products (laxatives, enemas)	**Accumulation**

11. Prevention: Sick Day Rules

If the patient becomes **dehydrated** (vomiting, diarrhea, fever), **temporarily hold** the following drugs.
Mnemonic: SADMANS

Letter	Drug
S	**Sulfonylureas**
A	**ACE inhibitors**
D	**Diuretics**
M	**Metformin**
A	**ARBs**
N	**NSAIDs**
S	**SGLT2 inhibitors**

12. Special Situations

Rhabdomyolysis

- **CK > 5 × upper limit of normal** (often > 5,000).
- **Tea- or cola-colored urine**.
- **Treatment: aggressive IV fluids** (target urine output **200–300 mL/h**).
- **Hyperkalemia and hyperphosphatemia**, with **hypocalcemia**.

Tumor lysis syndrome

- **Hyperkalemia, hyperphosphatemia, hyperuricemia**, with **hypocalcemia**.
- **Prevention:** hydration + **allopurinol** (start **2–3 days** before chemotherapy) or **rasburicase** for high-risk patients (**contraindicated in G6PD deficiency**, because it causes **hemolysis** and **methemoglobinemia**).

Exam Traps and Frequently Tested Points

1. **FENa < 1%** → **prerenal**. **> 2%** → **ATN**.
2. **Patient on diuretics** → use **FEUrea** (< 35% → prerenal).
3. **BUN:SCr > 20** → **prerenal**.
4. **Muddy brown casts** → **ATN**. **WBC casts + eosinophils** → **AIN**. **RBC casts** → **GN**.
5. **NSAIDs** → **afferent vasoconstriction**. **ACEi/ARB** → **efferent vasodilation**.
6. **NSAID + ACEi/ARB + diuretic** → **triple whammy**.
7. **Creatinine rise up to 30%** after ACEi → **acceptable**.
8. **Aminoglycosides** → after **5–7 days**, **non-oliguric**, **hypomagnesemia**.
9. **Amphotericin B** → **saline before and after the dose**, **hypokalemia and hypomagnesemia**.
10. **Cisplatin** → **hypomagnesemia**.
11. **Vancomycin + pip-tazo** → **more AKI**.
12. **PPIs and beta-lactams** → **AIN**.
13. **IV acyclovir** → **crystals**; prevent with **hydration and slow infusion**.
14. **Trimethoprim, cimetidine, dolutegravir** → **false rise in creatinine**.
15. **Contrast** → prevent with **saline**; **NAC is not beneficial**.
16. **Metformin + contrast + eGFR < 30** → **hold for 48 hours**.
17. **Loop diuretics** → **only for volume overload**; they **do not treat AKI**.
18. **Low-dose dopamine** → **not beneficial**.
19. **Hydroxyethyl starch (HES)** → **avoid**; it increases AKI.
20. **Hyperkalemia with ECG changes** → **calcium gluconate first**; it **does not lower potassium**.
21. **AEIOU** → dialysis indications.
22. **Loading dose is unchanged** in renal failure.
23. **Cockcroft-Gault is inaccurate** in AKI → assume **CrCl < 10** if creatinine is rising rapidly.
24. **Rasburicase + G6PD deficiency** → **contraindicated**.
25. **Meperidine** → **avoid** in renal failure.

> **Note:** Some cut-off values differ slightly between references, such as FENa, contrast prophylaxis doses, and the eGFR thresholds for metformin. **Use the numbers in the reference you are studying for the exam.**

Good luck on your exam!